View and update MetroCluster interconnects

ONTAP 9.11.1 REST API reference

NetApp
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View and update MetroCluster interconnects

Cluster MetroCluster interconnects endpoint overview

Overview

You can use this API to retrieve and display relevant information pertaining to MetroCluster interconnect status. The `/cluster/metrocluster/interconnects` endpoint returns a list of all the interconnects in MetroCluster and their status. Each individual interconnect can be queried individually using the `/cluster/metrocluster/interconnects/{node.uuid}/{partner_type}/{adapter}` endpoint. You can also use this API to modify relevant information related to MetroCluster interconnect. These include address, netmask, and gateway. Modify a MetroCluster interconnect using the `/cluster/metrocluster/interconnects/{node.uuid}/{partner_type}/{adapter}` endpoint.

Examples

Retrieving MetroCluster interconnects

```
GET https://<mgmt-ip>/api/cluster/metrocluster/interconnects
{
  "records": [
      {
          "node": {
              "name": "cluster1_01",
              "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
              "_links": {
                  "self": {
                      "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
005056826931"
                  }
              }
          },
          "partner_type": "ha",
          "adapter": "e0f",
          "_links": {
              "self": {
                  "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/ha/e0f"
              }
          }
      },
      {
          "node": {
              "name": "cluster1_01",
```

```
                "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
005056826931"
                    }
                }
            },
            "partner_type": "ha",
            "adapter": "e0g",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/ha/e0g"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_01",
                "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
005056826931"
                    }
                }
            },
            "partner_type": "dr",
            "adapter": "e0f",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/dr/e0f"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_01",
                "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
```

```
005056826931"
                    }
                }
            },
            "partner_type": "dr",
            "adapter": "e0g",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/dr/e0g"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_01",
                "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
005056826931"
                    }
                }
            },
            "partner_type": "aux",
            "adapter": "e0f",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/aux/e0f"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_01",
                "uuid": "6fead8fe-8d81-11e9-b5a9-005056826931",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/6fead8fe-8d81-11e9-b5a9-
005056826931"
                    }
                }
            },
```

```
            "partner_type": "aux",
            "adapter": "e0g",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/6fead8fe-8d81-11e9-b5a9-
005056826931/aux/e0g"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "ha",
            "adapter": "e0f",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/ha/e0f"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "ha",
            "adapter": "e0g",
            "_links": {
                "self": {
```

```
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/ha/e0g"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "dr",
            "adapter": "e0f",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/dr/e0f"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "dr",
            "adapter": "e0g",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/dr/e0g"
                }
```

```
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "aux",
            "adapter": "e0f",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/aux/e0f"
                }
            }
        },
        {
            "node": {
                "name": "cluster1_02",
                "uuid": "f5435191-8d81-11e9-9d4b-00505682dc8b",
                "_links": {
                    "self": {
                        "href": "/api/cluster/nodes/f5435191-8d81-11e9-9d4b-
00505682dc8b"
                    }
                }
            },
            "partner_type": "aux",
            "adapter": "e0g",
            "_links": {
                "self": {
                    "href":
"/api/cluster/metrocluster/interconnects/f5435191-8d81-11e9-9d4b-
00505682dc8b/aux/e0g"
                }
            }
        }
    ],
    "num_records": 12,
```

```
  "_links": {
      "self": {
          "href": "/api/cluster/metrocluster/interconnects"
      }
  }
}
```

Retrieves information about a specific MetroCluster interconnect

```
https://<mgmt-ip>/api/cluster/metrocluster/interconnects/774b4fbc-86f9-
11e9-9051-005056825c71/aux/e0f
{
  "node": {
      "name": "cluster1_01",
      "uuid": "46147363-9857-11e9-9a55-005056828eb9",
      "_links": {
          "self": {
              "href": "/api/cluster/nodes/46147363-9857-11e9-9a55-
005056828eb9"
          }
      }
  },
  "partner_type": "aux",
  "adapter": "e0f",
  "state": "up",
  "type": "iwarp",
  "interfaces": [
    {
      "address": "10.2.3.5",
      "netmask": "255.255.255.0"
    }
  ],
  "mirror": {
      "state": "online",
      "enabled": true
  },
  "multipath_policy": "static_map",
  "_links": {
      "self": {
          "href": "/api/cluster/metrocluster/interconnects/46147363-9857-
11e9-9a55-005056828eb9/ha/e0f"
      }
  }
}
```

This example shows how to modify the network address assigned to the home port. Fields required: address.

```
curl -X PATCH https://<mgmt-
ip>/api/cluster/metrocluster/interconnects/3e1bfd38-ffd2-11eb-bcb7-
005056aceaa9/ha/e0g -d '{"interfaces": [{"address": "1.2.3.4"}]}'
```

PATCH Response

```
HTTP/1.1 200 OK
Cache-Control: no-cache,no-store,must-revalidate
Connection: close
Date: Fri, 20 Aug 2021 21:58:36 GMT
Server: libzapid-httpd
Content-Length: 3
Content-Type: application/hal+json
X-Content-Type-Options: nosniff
{
}
```

This example shows how to modify the netmask assigned to the interface. Be sure to change to a valid subnet. Fields required: netmask.

```
curl -X PATCH https://<mgmt-
ip>/api/cluster/metrocluster/interconnects/3e1bfd38-ffd2-11eb-bcb7-
005056aceaa9/ha/e0g -d '{"interfaces": [{"netmask": "2.2.2.2"}]}'
```

PATCH Response

```
HTTP/1.1 200 OK
Cache-Control: no-cache,no-store,must-revalidate
Connection: close
Date: Fri, 20 Aug 2021 22:11:35 GMT
Server: libzapid-httpd
Content-Length: 3
Content-Type: application/hal+json
X-Content-Type-Options: nosniff
{
}
```

This example shows how to modify the gateway assigned to the interface. Please make sure to update it on the switch/router first. Assuming it is a new one, the new gateway and IP address must reside in the same subnet range as the interface IP address. Fields required: gateway.

```
curl -X PATCH https://<mgmt-
ip>/api/cluster/metrocluster/interconnects/3e1bfd38-ffd2-11eb-bcb7-
005056aceaa9/ha/e0g -d '{"interfaces": [{"gateway": "1.2.3.4"}]}'
```

PATCH Response

```
HTTP/1.1 200 OK
Cache-Control: no-cache,no-store,must-revalidate
Connection: close
Date: Fri, 20 Aug 2021 22:11:35 GMT
Server: libzapid-httpd
Content-Length: 3
Content-Type: application/hal+json
X-Content-Type-Options: nosniff
{
}
```

Retrieve interconnect adapter information for nodes in MetroCluster

GET `/cluster/metrocluster/interconnects`

Introduced In: 9.8

Retrieves a list of interconnect adapter information for nodes in the MetroCluster.

Related ONTAP Commands

- `metrocluster interconnect show`

Learn more

- DOC /cluster/metrocluster/interconnects

Parameters

Name	Type	In	Required	Description
partner_type	string	query	False	Filter by partner_type
adapter	string	query	False	Filter by adapter

Name	Type	In	Required	Description
vlan_id	integer	query	False	Filter by vlan_id • Introduced in: 9.9 • Max value: 4095 • Min value: 10
type	string	query	False	Filter by type
multipath_policy	string	query	False	Filter by multipath_policy • Introduced in: 9.11
mirror.state	string	query	False	Filter by mirror.state • Introduced in: 9.11
mirror.enabled	boolean	query	False	Filter by mirror.enabled • Introduced in: 9.11
state	string	query	False	Filter by state
node.uuid	string	query	False	Filter by node.uuid
node.name	string	query	False	Filter by node.name
interfaces.gateway	string	query	False	Filter by interfaces.gateway • Introduced in: 9.9
interfaces.netmask	string	query	False	Filter by interfaces.netmask • Introduced in: 9.9

Name	Type	In	Required	Description
interfaces.address	string	query	False	Filter by interfaces.address • Introduced in: 9.9
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Max value: 120 • Min value: 0 • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	collection_links	
num_records	integer	Number of Records
records	array[metrocluster_interconnect]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "adapter": "string",
    "interfaces": {
      "address": "10.10.10.7",
      "gateway": "10.1.1.1",
      "netmask": "24"
    },
    "mirror": {
      "state": "online"
    },
    "multipath_policy": "no_mp",
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "partner_type": "aux",
    "state": "down",
    "type": "roce",
    "vlan_id": 0
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
2425734	An internal error occurred. Wait a few minutes, and try the operation again. For further assistance, contact technical support.
2427132	MetroCluster is not configured on this cluster.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

collection_links

Name	Type	Description
next	href	
self	href	

self_link

Name	Type	Description
self	href	

interfaces

Object to setup an interface along with its default router.

Name	Type	Description
address	string	IPv4 or IPv6 address
gateway	string	The IPv4 or IPv6 address of the default router.
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length.

mirror

Name	Type	Description
enabled	boolean	Specifies the administrative state of the NVRAM mirror between partner nodes.
state	string	Specifies the operational state of the NVRAM mirror between partner nodes.

_links

Name	Type	Description
self	href	

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

metrocluster_interconnect

Data for a MetroCluster interconnect. REST: /api/cluster/metrocluster/interconnects

Name	Type	Description
_links	self_link	
adapter	string	Adapter
interfaces	array[interfaces]	List of objects which contain interface information such as its IP address, netmask and gateway.
mirror	mirror	
multipath_policy	string	Displays the NVRAM mirror multipath policy for the nodes configured in a MetroCluster.
node	node	
partner_type	string	Partner type
state	string	Adapter status
type	string	Adapter type
vlan_id	integer	VLAN ID

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve information about a MetroCluster interconnect for a partner type and adapter

GET `/cluster/metrocluster/interconnects/{node.uuid}/{partner_type}/{adapter}`

Introduced In: 9.8

Retrieves information about a MetroCluster Interconnect for a specific partner type and adapter.

Related ONTAP Commands

- `metrocluster interconnect show`

Parameters

Name	Type	In	Required	Description
node.uuid	string	path	True	Node UUID
partner_type	string	path	True	DR Partner type
adapter	string	path	True	Interconnect adapter.
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	self_link	
adapter	string	Adapter
interfaces	array[interfaces]	List of objects which contain interface information such as its IP address, netmask and gateway.
mirror	mirror	
multipath_policy	string	Displays the NVRAM mirror multipath policy for the nodes configured in a MetroCluster.
node	node	
partner_type	string	Partner type
state	string	Adapter status
type	string	Adapter type
vlan_id	integer	VLAN ID

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "adapter": "string",
  "interfaces": {
    "address": "10.10.10.7",
    "gateway": "10.1.1.1",
    "netmask": "24"
  },
  "mirror": {
    "state": "online"
  },
  "multipath_policy": "no_mp",
  "node": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "node1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "partner_type": "aux",
  "state": "down",
  "type": "roce",
  "vlan_id": 0
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
2425734	An internal error occurred. Wait a few minutes, and try the operation again. For further assistance, contact technical support.

Error Code	Description
2427132	MetroCluster is not configured on this cluster.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

self_link

Name	Type	Description
self	href	

interfaces

Object to setup an interface along with its default router.

Name	Type	Description
address	string	IPv4 or IPv6 address
gateway	string	The IPv4 or IPv6 address of the default router.
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length.

mirror

Name	Type	Description
enabled	boolean	Specifies the administrative state of the NVRAM mirror between partner nodes.
state	string	Specifies the operational state of the NVRAM mirror between partner nodes.

_links

Name	Type	Description
self	href	

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update a MetroCluster interconnect interface

PATCH `/cluster/metrocluster/interconnects/{node.uuid}/{partner_type}/{adapter}`

Introduced In: 9.10

Updates a MetroCluster interconnect interface.

Related ONTAP commands * `metrocluster configuration-settings interface modify`

Parameters

Name	Type	In	Required	Description
node.uuid	string	path	True	Node UUID
partner_type	string	path	True	DR Partner type

Name	Type	In	Required	Description
adapter	string	path	True	Interconnect adapter
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Request Body

Name	Type	Description
_links	self_link	
adapter	string	Adapter
interfaces	array[interfaces]	List of objects which contain interface information such as its IP address, netmask and gateway.
mirror	mirror	

Name	Type	Description
multipath_policy	string	Displays the NVRAM mirror multipath policy for the nodes configured in a MetroCluster.
node	node	
partner_type	string	Partner type
state	string	Adapter status
type	string	Adapter type
vlan_id	integer	VLAN ID

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "adapter": "string",
  "interfaces": {
    "address": "10.10.10.7",
    "gateway": "10.1.1.1",
    "netmask": "24"
  },
  "mirror": {
    "state": "online"
  },
  "multipath_policy": "no_mp",
  "node": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "node1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "partner_type": "aux",
  "state": "down",
  "type": "roce",
  "vlan_id": 0
}
```

Response

```
Status: 202, Accepted
```

Name	Type	Description
job	job_link	

Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
2425734	An internal error has occurred. Wait a few minutes, and try the operation again. For further assistance, contact technical support.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

self_link

Name	Type	Description
self	href	

interfaces

Object to setup an interface along with its default router.

Name	Type	Description
address	string	IPv4 or IPv6 address
gateway	string	The IPv4 or IPv6 address of the default router.
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length.

mirror

Name	Type	Description
enabled	boolean	Specifies the administrative state of the NVRAM mirror between partner nodes.
state	string	Specifies the operational state of the NVRAM mirror between partner nodes.

_links

Name	Type	Description
self	href	

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

metrocluster_interconnect

Data for a MetroCluster interconnect. REST: /api/cluster/metrocluster/interconnects

Name	Type	Description
_links	self_link	
adapter	string	Adapter
interfaces	array[interfaces]	List of objects which contain interface information such as its IP address, netmask and gateway.
mirror	mirror	
multipath_policy	string	Displays the NVRAM mirror multipath policy for the nodes configured in a MetroCluster.
node	node	
partner_type	string	Partner type
state	string	Adapter status
type	string	Adapter type
vlan_id	integer	VLAN ID

job_link

Name	Type	Description
_links	_links	
uuid	string	The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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